Preparation of a software project plan covering scope, timeline, and team roles for a sample application

Aim :

To prepare a comprehensive software project plan for a Spotify-like music streaming application by defining the project scope, objectives, timeline, team roles, deliverables, and resource allocation to ensure systematic planning and successful project execution.

Objective:

The objective of this project is to prepare a software project plan for a Spotify-like music streaming application by defining its scope, timeline, team roles, resources, and key deliverables. The plan provides a structured roadmap to ensure efficient project planning, execution, and successful completion.

Selected Software Application

Spotify-like Music Streaming Application

A Spotify-like Music Streaming Application allows users to:

- Register/Login securely
- Search for songs, albums, artists, and playlists
- Stream music online
- Create and manage personal playlists
- Like and save favorite songs
- Follow artists and receive updates
- Download songs for offline listening (Premium users)
- Receive personalized music recommendations
- Share songs and playlists with others
- Manage subscription plans (Free and Premium)
- View listening history and recently played tracks

This application also provides artists with the ability to upload and manage their music, while administrators can manage users, music content, subscriptions, and platform operations.

Functional Requirements :

User Module

- Sign Up / Login
- Search songs
- Browse albums and artists
- Stream music
- Create playlists
- Like/Favorite songs
- Download songs (Premium)
- Share playlists
- View listening history
- Manage profile

Artist Module

- Upload songs
- Manage albums
- View song analytics
- Edit artist profile

Admin Module

- Manage users
- Manage artists
- Approve uploaded songs
- Remove inappropriate content
- Generate reports
- Manage subscriptions

Scope of the application:

The Spotify-like Music Streaming Application is designed to provide users with a seamless and personalized music streaming experience.

- It enables users to register, log in, search for songs, albums, artists, and playlists, stream music online, create and manage playlists, save favorite tracks, and receive personalized recommendations.
- Premium users can also download songs for offline listening and enjoy an ad-free experience. The application includes dedicated modules for artists to upload and manage their music and for administrators to oversee users, content, and subscription plans.
- The project scope focuses on delivering a secure, scalable, and user-friendly platform that supports efficient music streaming across web and mobile devices.

In Scope

- User registration and authentication
- Music search and streaming
- Playlist creation and management
- Favorite songs and listening history
- Artist profiles and music uploads
- Premium subscription and offline downloads
- Admin dashboard for user and content management

Out of Scope

- Live concerts and event streaming
- Podcast production and publishing
- Music editing or recording tools
- Video streaming
- Social media messaging features

Team Role :

Team Members	Role	Responsibility
Project Manager	Project Planning	Schedule, budget, coordination
Business Analyst	Requirement Analysis	Gather and document requirements
UI/UX Designer	Interface Design	Design app screens and user experience
Front-End Developer	App Development	Build Android, iOS, and web interfaces
Back-End Developer	Server Development	Develop APIs, authentication, streaming services
Database Administrator	Database Management	Design and maintain the database
QA/Test Engineer	Testing	Functional, integration, and performance testing
DevOps Engineer	Deployment	CI/CD, cloud deployment, monitoring

Risk Management:

Risk	**Impact**	**Mitigation**
Server overload	High	Auto-scaling cloud infrastructure
Copyright issues	High	Verify licensing before publishing music
Security breaches	High	Encryption, MFA, regular security audits
Internet connectivity	Medium	Offline downloads for Premium users
Requirement changes	Medium	Agile development with regular reviews

Work Breakdown Structure (WBS):

